

Carlos R Moraes Rodrigues

Senior Software & DevOps Engineer

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PROFESSIONAL SUMMARY

Software engineer with more than 20 years of experience, combining product development and critical systems operations. Develops APIs and integrations in Go, Python and PHP, Vue interfaces and data pipelines, connecting business rules, testing and delivery. Works with AWS, Docker Swarm, IaC, CI/CD, observability and FinOps. Applies specification driven development and AI with human review, investigating issues across applications, Linux, networking and infrastructure.

TECHNICAL SKILLS

Software: Go, Python, FastAPI, PHP, Laravel, Symfony, Vue 3, Quasar, REST APIs

Data and integrations: PostgreSQL, SQL Server, RabbitMQ, Elasticsearch, Logstash, SIP/VoIP

Cloud and containers: AWS, EC2, RDS, S3, VPC, IAM, ALB, ACM, EFS, Docker Swarm, Traefik

Infrastructure as Code (IaC) and automation: Terraform, Ansible, Bash

CI/CD and quality: GitHub Actions, Bitbucket Pipelines, unit and integration testing, k6

Observability and FinOps: Grafana, Prometheus, Loki, Alloy

Practices: spec-driven development, Codex, Figma/MCP, Platform Engineering, SRE foundations

PROFESSIONAL EXPERIENCE

Senior Developer - Software, DevOps and Platform Engineering

Seventh | Apr 2021 - Present | Palhoça, SC, Brazil

- Developed a Go API for video surveillance sizing and calibration, with Vue 3/Quasar interfaces, REST contracts, tests and internationalization.
- Evolved Python/FastAPI integrations across FreeSWITCH, OpenSIPS and RabbitMQ; built a VoIP lab using SIP/RTP to reproduce calls and validate workflows.
- Developed a Go/PostgreSQL API for SIP snapshots and validated 750 individual requests/s for 5 min after warm-up with k6, with p95 of 184 ms and zero HTTP errors.
- Applied specification driven development and created an engineering playbook with Codex, Figma/MCP, acceptance criteria, tests and human approval.
- Built and maintains an AWS/Docker Swarm platform with ALB, ACM and Traefik across three Availability Zones, validating the change in staging reproduced with Terraform.
- Modernized legacy PHP into containers with PHP-FPM, Nginx, WAF, queues and schedulers; automated deployments and operations with Bitbucket Pipelines, Python and Bash.
- Implemented Grafana, Prometheus, Loki and Alloy observability; reduced unwanted inter-AZ traffic by over 99% and classified more than 100 AWS resources for FinOps.
- Developed and operated Logstash/SQL Server pipelines for BI and billing, with Elasticsearch mappings, incremental loads, transformations and Kibana dashboards.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior PHP Developer

Conectaa Sistemas | Aug 2014 - Feb 2021 | Florianópolis, SC, Brazil

- Developed and maintained corporate web applications with PHP, Laravel and Drupal, focusing on integrations, legacy system evolution and institutional applications.
- Built integrations between corporate systems and portals through APIs and services, also working in Linux environments and supporting critical applications.
- Delivered solutions for legislative institutions in Santa Catarina, Bahia and Florianópolis.

Freelance Web Developer

Self-employed | 2011 - 2014

Developed and maintained web applications with PHP, JavaScript, relational databases and Linux, balancing independent projects with studies in Clean Code, Design Patterns, architecture and automated testing.

Mid-level PHP Developer

Fundação CEEE de Seguridade Social | Nov 2007 - Mar 2011 | Porto Alegre, RS, Brazil

Developed and maintained internal corporate and self-service systems with PHP, CodeIgniter and PostgreSQL for operational and administrative teams.

Systems Developer

Sapient AG2 | Jan 2003 - Nov 2007 | Porto Alegre, RS, Brazil

Developed web applications and corporate systems with Microsoft .NET and PHP for clients in the financial, industrial and retail sectors.

Early software development experience

Fundação Educacional and Data Bank do Brasil | 2001 - 2003 | RS, Brazil

Developed desktop systems, library solutions, databases and integrations with electronic devices.

SELECTED PROJECTS

aws-swarm-lab - Highly available Docker Swarm on AWS

Public, reproducible infrastructure-as-code lab using Terraform, Ansible, Docker Swarm, EC2, EFS, ALB, ACM and Traefik. github.com/robmoraes/aws-swarm-lab

Distributed Systems Lab - Original DevOps technical content

Educational channel organized into progressive series. The first published season has 10 episodes and manually builds the foundation of a Docker Swarm lab on AWS, progressing from EC2 and port-based exposure to a stack with WAF, a domain, Traefik and TLS. Includes a public playlist and a companion quiz produced with Quick Quiz, whose administration tool uses OpenAI to generate and translate questions on demand.

Pinleaf - Linux desktop application

Open-source Python application with CI, tests, GitHub Pages, Debian packaging and automated releases through GitHub Actions. github.com/robmoraes/pinleaf

EDUCATION

Higher Education Technology Program in Data Processing - Universidade Católica de Pelotas | 1998-2000

DevOps Ninja: Docker, Kubernetes and Rancher - Udemy | Jan 2022 | 18 hours